

A Machine-Checked Finite Reduction for Erdős Problem #287 and a Source-Explicit Large- M Criterion

Clean Public Review – Version 16

Research draft with Lean/Aristotle-assisted formal verification

28 August 2026

Status. Erdős Problem #287 remains open. Public repository `main` machine-checks the exact problem statement, the finite exclusion through maximum denominator 4×10^9 , the conditional window-pair closure compiler, and the V14 finite/algebraic Vaughan source spine. The later V15 $\Lambda = \mu * \log$ and balanced-seven Lean files are present in the supplied midpoint Aristotle archive, but were not yet on public `main` at this audit cutoff and were not independently rebuilt in the present execution environment. They are therefore labelled *formalization source present – public replay pending*, not public Lean theorems. No prime-distribution estimate, Ford–Maynard lower bound, or final large- M supply is claimed proved. V16 source commit: `85f8b47cee65`.

Abstract

Erdős Problem #287 asks whether every representation

$$1 = \frac{1}{n_1} + \cdots + \frac{1}{n_k}, \quad 1 < n_1 < \cdots < n_k,$$

contains a consecutive gap of size at least 3. This note separates the present work into three logically distinct layers.

First, the finite layer is machine-checked: there is no counterexample whose maximum denominator is at most 4×10^9 . The proof uses a prime-power window obstruction, an adjacent-hole contradiction, and a chain of interval certificates rather than a search over all unit-fraction representations.

Second, the large- M conclusion is compiled from one explicit window-pair supply predicate. It asks for two consecutive integers in the top half of $[1, M]$, each carrying a sufficiently large prime-power divisor. A Lean theorem shows that an effective threshold $M_0 \leq 4 \times 10^9$ together with `WindowPairSupply( $M$ )` for every $M \geq M_0$ implies the exact public statement. No inhabitant of this large- M predicate is presently known.

Third, the current analytic research is made source-explicit. The public V14 bank contains an exact Vaughan source spine. A supplied midpoint V15 formalization develops the source-minimal identity $\Lambda = \mu * \log$, its affine specialization at $2mn \pm 1$, an exact three-way factor partition, a determinant-one line parametrization, and finite balanced-seven polarization algebra. Those V15 files are treated here as formalization source pending final public replay; the mathematical identities are displayed explicitly and do not depend on an analytic estimate. The remaining analytic statements are a signed endpoint distribution estimate for a polarized family of $X^{1/7}$ -smooth multiplicative functions at moduli $X^{3/5}$, its comparison/low-conductor matching statement, and the remaining source-minimal hard-line estimates. These are open specifications, not theorems.

Status at a glance

Item	Status	Meaning
Exact public counterexample predicate	PUBLIC LEAN-CHECKED	Set form and ordered sequence form are connected without adding a hidden $k \geq 2$ hypothesis.
Finite range $M \leq 4 \times 10^9$	PUBLIC LEAN-CHECKED	A 34-certificate interval chain excludes every exact counterexample in this range.
Window-pair closure compiler	PUBLIC LEAN-CHECKED, CONDITIONAL	An explicit large- M supply predicate implies the full conjecture when paired with the finite range.
V14 Vaughan affine source spine	PUBLIC LEAN-CHECKED ALGEBRA	Exact decomposition and structural compilers; analytic inputs remain uninhabited.
$\Lambda = \mu * \log$, hard-pair line, balanced-seven algebra	FORMALIZATION REPLAY PENDING	V15 source is present in the midpoint archive and statically audited; it was not yet replayed on public <code>main</code> at cutoff.
Large- M prime distribution	OPEN	No proof of <code>WindowPairSupply(M)</code> for every sufficiently large M .
Polarized signed endpoint estimate	OPEN ANALYTIC	Current exact V-branch residual at $(Q, y) = (X^{3/5}, X^{1/7})$.
Comparison/low-conductor matching	OPEN SOURCE	The physical comparison sequence must undergo the same decomposition.
Erdős Problem #287	OPEN	No unconditional proof is claimed.

Status vocabulary. PUBLIC LEAN-CHECKED means the declaration is on public repository `main`. EXACT ALGEBRA means the identity is proved in the exposition. FORMALIZATION REPLAY PENDING means Lean source is present in the supplied midpoint archive but has not yet been promoted into the public bank. CONDITIONAL means a compiler is proved from an explicit antecedent. OPEN means that antecedent or analytic estimate is not proved. FALSE/RETIRED is reserved for a literal failed dictionary or superseded route, never for the underlying general theorem.

Contents

Status at a glance	2
1 The problem and the scope of this note	3
2 The machine-checked finite theorem	3
2.1 Holes and the adjacent-hole contradiction	4
2.2 Prime-power windows	4
2.3 The interval-certificate chain	4
3 The exact large-M closure criterion	5
4 The source-minimal affine algebra	5
4.1 The identity $\Lambda = \mu * \log$	5
4.2 Exact factor partition	6
4.3 The determinant-one line	6

5	The balanced-seven polarization	6
5.1	Why separate character energies do not close	7
5.2	Exact squarefree polarization	7
5.3	The exact V-branch decomposition	7
6	What existing distribution theorems do and do not supply	8
7	The current end-to-end dependency graph	9
8	Formal verification and reproducibility	9
8.1	Public Lean bank	9
8.2	Midpoint V15 archive	9
8.3	Document organization	10
9	Conclusion	10
A	Formal theorem and source map	10

1 The problem and the scope of this note

The public problem asks whether, for distinct integers

$$1 < n_1 < \cdots < n_k, \quad \sum_{i=1}^k \frac{1}{n_i} = 1,$$

one necessarily has

$$\max_{1 \leq i < k} (n_{i+1} - n_i) \geq 3.$$

It is listed as open in the Erdős Problems archive and originates in the sources cited there [1, 2].

For a finite set $A \subset \mathbb{N}$, write

$$N = \min A, \quad M = \max A.$$

A counterexample is a finite set satisfying

$$A \subset \{2, 3, \dots\}, \quad \sum_{a \in A} \frac{1}{a} = 1,$$

and the local gap condition that every nonmaximal $a \in A$ is followed by $a + 1$ or $a + 2$ in A . The Lean development formalizes exactly this predicate and proves the ordered form from it.

This note is deliberately not a chronological research diary. It records only the current controlling finite theorem, the exact conditional closure interface, the source algebra with its precise public/formalization status, and the current open analytic obligations. Earlier provider searches, hostile audits, and superseded residuals are retained in the separate technical verification ledger and immutable V15 archive.

2 The machine-checked finite theorem

Theorem 2.1 (Finite exclusion). *There is no exact counterexample to Erdős Problem #287 with*

$$3 \leq M \leq 4,000,000,000.$$

Equivalently, every unit-fraction representation in the problem with maximum denominator at most 4×10^9 has a consecutive gap at least 3.

The theorem is represented in the repository by `no_Erdos287Counterexample_of_max_le_4e9`. Its proof is not an exhaustive enumeration of finite subsets. It is a certificate compiler built from the three ideas below.

2.1 Holes and the adjacent-hole contradiction

For a hypothetical counterexample, call an integer $x \in [N, M]$ a *hole* when $x \notin A$. The $\text{gap} \leq 2$ condition implies that two consecutive integers in $[N, M]$ cannot both be holes. This is the formal theorem `Gap2CE.holes_isolated`.

A second elementary fact places the top half of the interval above the minimum:

$$N \leq \left\lfloor \frac{M}{2} \right\rfloor.$$

Thus, whenever $M \leq 2x$ and $x + 1 \leq M$, the consecutive pair $x, x + 1$ lies inside the closed window where the adjacent-hole contradiction applies.

2.2 Prime-power windows

For $j \geq 0$, let $C(j)$ denote the largest reduced numerator arising from a nonempty partial reciprocal sum

$$\sum_{s \in S} \frac{1}{s}, \quad \emptyset \neq S \subseteq \{1, \dots, j\}.$$

The formal development uses a certified upper table

$$C_{\text{val}}(0), \dots, C_{\text{val}}(9) = 1, 1, 3, 11, 25, 137, 137, 1019, 2143, 7129.$$

The prime-power window theorem says, schematically, that if p is prime, $e \geq 1$, $p^e \mid x$, and

$$C\left(\left\lfloor \frac{M}{p^e} \right\rfloor\right) < p,$$

then x cannot belong to A . The reason is that the top p -adic layer of the reciprocal sum would force a nonzero numerator to vanish modulo p .

Consequently, if both x and $x + 1$ possess such prime-power divisors, then they are two adjacent holes. This gives the central finite compiler.

Proposition 2.2 (Window-pair blocker). *Suppose there are primes p_u, p_v , positive exponents a_u, a_v , and an integer x such that*

$$\begin{aligned} p_u^{a_u} \mid x, \quad p_v^{a_v} \mid x + 1, \\ \left\lfloor \frac{M}{p_u^{a_u}} \right\rfloor, \left\lfloor \frac{M}{p_v^{a_v}} \right\rfloor \leq 9, \\ C_{\text{val}}\left(\left\lfloor \frac{M}{p_u^{a_u}} \right\rfloor\right) < p_u, \quad C_{\text{val}}\left(\left\lfloor \frac{M}{p_v^{a_v}} \right\rfloor\right) < p_v, \end{aligned}$$

and

$$M \leq 2x, \quad x + 1 \leq M.$$

Then a $\text{gap} \leq 2$ counterexample with maximum M cannot exist.

2.3 The interval-certificate chain

One certificate can exclude an entire interval $L \leq M \leq U$. The divisibility data are fixed at a chosen x , monotonicity controls the windows as M varies, and $U \leq 2x$ keeps the forced holes above the minimum. The repository contains a chain of 34 certificates whose intervals meet end-to-end and cover

$$[3, 4 \times 10^9].$$

All primality side conditions are discharged in the kernel; the construction does not use `native_decide`.

3 The exact large- M closure criterion

Definition 3.1 (Window-pair supply). For $M \in \mathbb{N}$, $\text{WindowPairSupply}(M)$ is the assertion that there exist $x, p_u, a_u, p_v, a_v \in \mathbb{N}$ such that

$$\begin{aligned} p_u, p_v &\text{ are prime,} & a_u, a_v &\geq 1, \\ p_u^{a_u} &\mid x, & p_v^{a_v} &\mid x+1, \\ \left\lfloor \frac{M}{p_u^{a_u}} \right\rfloor &\leq 9, & C_{\text{val}}\left(\left\lfloor \frac{M}{p_u^{a_u}} \right\rfloor\right) &< p_u, \\ \left\lfloor \frac{M}{p_v^{a_v}} \right\rfloor &\leq 9, & C_{\text{val}}\left(\left\lfloor \frac{M}{p_v^{a_v}} \right\rfloor\right) &< p_v, \\ M &\leq 2x, & x+1 &\leq M. \end{aligned}$$

In plain terms, two consecutive integers in the top half of $[1, M]$ each have a prime-power divisor larger than roughly $M/10$, with a small local numerator condition.

Theorem 3.2 (End-to-end conditional compiler). *Assume there is an explicit $M_0 \leq 4 \times 10^9$ such that*

$$\text{WindowPairSupply}(M) \quad \text{for every } M \geq M_0.$$

Then Erdős Problem #287 is true.

This implication is machine-checked as `no_Erdos287Counterexample_of_closure`. The proof splits at 4×10^9 : the finite theorem handles the lower range, while the window-pair blocker handles every larger maximum.

The interface is weaker than a Sophie-Germain prime condition. For example, if a prime q in the interval $M/3 < q \leq (M-1)/2$ has $2q+1$ prime, then one may take $x = 2q$. The analogous minus case uses $x = 2q-1$. The Lean theorem `windowPairSupply_of_sophieWitness` formalizes this implication. The present programme does not claim such Sophie witnesses for every large M .

4 The source-minimal affine algebra

This section concerns the analytic route toward a large- M supply. The formulas below are exact and are proved algebraically in this note. Matching V15 Lean source is present in the supplied midpoint archive, but final public replay was pending at the audit cutoff; accordingly this section is not labelled public Lean-checked. All cancellation estimates remain open.

4.1 The identity $\Lambda = \mu * \log$

For the genuine arithmetic functions, Dirichlet convolution gives

$$\mu * \log = \mu * (\Lambda * 1) = (\mu * 1) * \Lambda = \Lambda,$$

and hence

$$\Lambda(N) = \sum_{qr=N} \mu(q) \log r.$$

Equivalently,

$$\Lambda(N) = \sum_{q \mid N} \mu(q) \log \frac{N}{q}.$$

The proof is carried out in the Dirichlet convolution ring from $\mu * 1 = \varepsilon$ and $\log = \Lambda * 1$; it agrees definitionally with Mathlib's own identity.

At the affine value

$$N = 2mn + s, \quad s \in \{-1, +1\},$$

one obtains the exact source

$$\Lambda(2mn + s) = \sum_{qr=2mn+s} \mu(q) \log r.$$

The sign is represented by a dedicated formal datatype, preventing illegal subtraction in $\mathbb{N}$ for the case $s = -1$.

4.2 Exact factor partition

For a cutoff U , the factor pairs are partitioned disjointly into

$$q \leq U, \quad q > U, r \leq U, \quad q > U, r > U.$$

The three classes are exhaustive and pairwise disjoint, so an arbitrary weighted sum recombines exactly over them. The hard part is therefore

$$\sum_{\substack{qr=2mn+s \\ q>U, r>U}} \mu(q) \log r.$$

No estimate for this sum is encoded.

4.3 The determinant-one line

Every hard pair satisfies

$$qr - 2mn = s, \quad s = \pm 1.$$

This unit shift forces

$$(r, 2m) = 1, \quad (q, 2n) = 1.$$

Fixing r, m and one solution (q_0, n_0) , all integer solutions are exactly

$$\boxed{q = q_0 + 2mt, \quad n = n_0 + rt, \quad t \in \mathbb{Z},}$$

and the parameter t is unique. The forward direction, reverse direction, coprimality, nonvanishing, and multiplicity one are elementary consequences of the unit determinant; matching V15 Lean declarations are present in the midpoint archive pending public replay.

The exact exponent ledger records, for the programme parameter $\sigma < 1/6$,

$$\sigma + \frac{2}{3} + \frac{1}{6} < 1, \quad \frac{1}{6} - \frac{16623}{100000} = \frac{131}{300000} > 0.$$

These are exponent identities only; they are not bounds for an analytic sum.

Remark 4.1 (Relation to the V14 Vaughan route). The earlier exact Vaughan decomposition and prime-modulus Möbius two-outer packet remain valid alternative source decompositions. They are not false, but the identity $\Lambda = \mu * \log$ is source-minimal and does not require truncation parameters or an additional convolution with 1. The clean public paper therefore treats the Vaughan packet as an optional analytic child, not as the unique controlling source.

5 The balanced-seven polarization

One explicit V-branch cell has seven labelled prime boxes. Put

$$Q = X^{3/5}, \quad Y = X^{1/7},$$

and consider

$$\mathcal{B}_{7,s}(X) = \sum_{q \sim Q} \mu(q) \left[\sum_{\substack{p_1, \dots, p_7 \sim Y \\ 2p_1 \cdots p_7 \equiv -s \pmod{q}}} \prod_{i=1}^7 \omega_i(p_i) - \mathfrak{M}_{7,s}(q) \right].$$

The desired scale is $o(X/\log X)$.

5.1 Why separate character energies do not close

Character orthogonality produces a sevenfold character product. Splitting it $3 + 4$ and using the ordinary multiplicative large sieve gives the scale

$$Q(X^{3/7}X^{4/7})^{1/2} = X^{3/5+1/2} = X^{11/10},$$

a deficit of $X^{1/10}$. Separate high-moment interpolation gives the same scale. This explains why the current reduction does not simply repeat a standard large-sieve argument.

5.2 Exact squarefree polarization

For $\mathbf{z} = (z_1, \dots, z_7) \in \mathbb{T}^7$, define

$$a_{\mathbf{z}}(p) = \frac{1}{7} \sum_{i=1}^7 z_i \omega_i(p).$$

Extend this to a squarefree multiplicative function $f_{\mathbf{z}}$ supported on primes in the box $p \sim Y$ by

$$f_{\mathbf{z}}(p) = a_{\mathbf{z}}(p), \quad f_{\mathbf{z}}(p^\ell) = 0 \quad (\ell \geq 2).$$

Expanding the polynomial shows that the coefficient of $z_1 \cdots z_7$ in

$$\prod_{j=1}^7 \left(\sum_{i=1}^7 z_i \omega_i(p_j) \right)$$

is exactly

$$\sum_{\sigma \in S_7} \prod_{j=1}^7 \omega_{\sigma(j)}(p_j),$$

with each permutation appearing once. The normalization factor is $7^{-7} = 1/823543$, and $|S_7| = 5040$.

Repeated-prime tuples have only six free prime variables, hence contribute at most

$$X^{6/7+o(1)},$$

which is below $X/\log^A X$ for every fixed A .

The same source audit gives the finite squarefree encoding and the binomial certificate

$$\sum_{j=0}^3 (-1)^j \binom{7}{j} = -20.$$

It does not identify this number with a physical Ford weight; the statement is purely finite combinatorics.

5.3 The exact V-branch decomposition

Let $\Delta_{\text{LC}}(f_{\mathbf{z}}; X, q, a_s)$ denote the progression discrepancy after removing the chosen principal and low-conductor terms, where $a_s \equiv -s^2 \pmod{q}$. Coefficient extraction gives

$$\begin{aligned} \mathcal{B}_{7,s}(X) &= 7^7 \int_{\mathbb{T}^7} \overline{z_1 \cdots z_7} \sum_{q \sim X^{3/5}} \mu(q) \Delta_{\text{LC}}(f_{\mathbf{z}}; X, q, a_s) d\mathbf{z} \\ &\quad + \sum_{q \sim Q} \mu(q) (\mathfrak{M}_{7,s}^{\text{pol}}(q) - \mathfrak{M}_{7,s}(q)) + O(X^{6/7+o(1)}). \end{aligned}$$

Thus this cell is reduced to two source-exact statements.

Polarized signed endpoint estimate.

$$\int_{\mathbb{T}^7} \overline{z_1 \cdots z_7} \sum_{q \sim X^{3/5}} \mu(q) \Delta_{\text{LC}}(f_{\mathbf{z}}; X, q, a_s) d\mathbf{z} = o(X/\log X).$$

Programme label: `AFFINE287-POLARIZED-OMEGA7-SIGNED-EOD45`.

Comparison/low-conductor match.

$$\sum_{q \sim Q} \mu(q) (\mathfrak{M}_{7,s}^{\text{pol}}(q) - \mathfrak{M}_{7,s}(q)) = o(X/\log X).$$

Programme label: `AFFINE287-MULOG-COMPARISON-LOWCOND-MATCH45`.

Both are open. The supplied V15 status source defines uninhabited specification structures for them, not proofs; that status file was not yet part of public `main` at the audit cutoff.

6 What existing distribution theorems do and do not supply

Ford and Maynard provide a general prime-producing sieve framework from Type I and Type II information for an actual nonnegative sequence [3]. Their theorem is a downstream framework, not a proof of the specific affine correlations above.

Drappeau, Granville and Shao prove distribution of smooth-supported multiplicative functions beyond the square-root barrier, up to moduli $x^{3/5-\varepsilon}$ under their stated hypotheses [4]. Pascadi obtains stronger exponents of distribution for primes and smooth numbers in a different weighted setting [5]. These results are close in scale to the polarized cell, but the current audit has not established a literal specialization at the endpoint

$$(Q, y) = (X^{3/5}, X^{1/7})$$

with the signed Möbius modulus average, torus coefficient, low-conductor subtraction, and physical comparison term all preserved. This note therefore records a promising neighbourhood of existing technology, not a black-box closure.

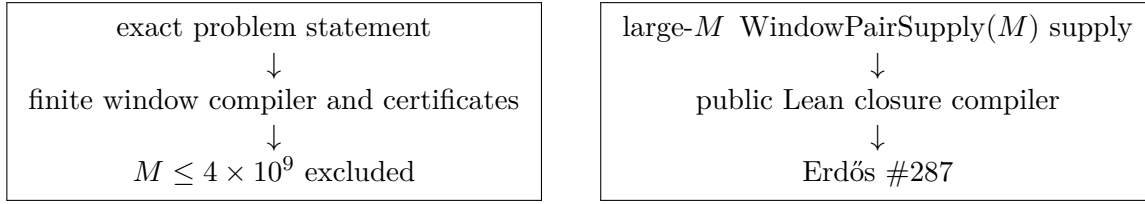
Several negative conclusions are also retained:

- the ordinary separate multiplicative large sieve misses by $X^{1/10}$;
- a three-block decomposition satisfying the relevant asymmetric ranges is unavailable for seven equal $1/7$ exponents;
- a termwise prime decomposition cannot remove every unsplit prime-box term;
- a modulus micro-switch cannot be exhaustive because prime moduli survive;
- the V14 truncated Möbius cofactor is not literally a well-factorable modulus weight;
- the canonical Gate-1A determinant- $2k$ row does not directly match a determinant ± 1 affine source.

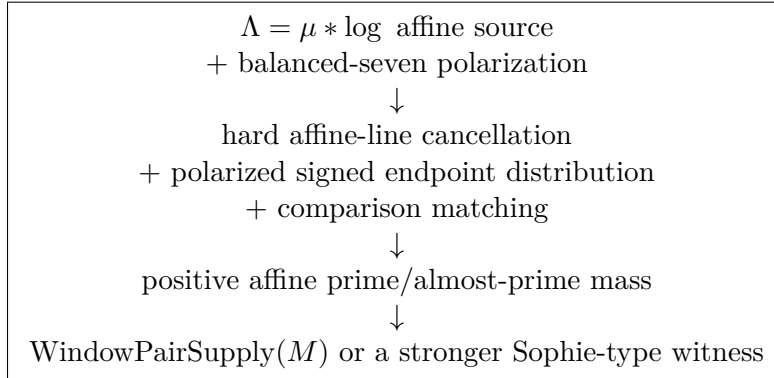
The detailed proofs and route-by-route provenance belong in the technical ledger, not in the main paper.

7 The current end-to-end dependency graph

The verified and open layers can now be displayed without historical notation:



The analytic programme seeks to produce the missing large- M supply. Its current source-explicit front is



Every downward arrow in the second display contains open analysis. The diagram is a research programme, not a proof. Moreover, closing the displayed balanced-seven cell does not by itself close all remaining fixed-certificate packets or the full large- M supply.

8 Formal verification and reproducibility

8.1 Public Lean bank

At V16 source commit 85f8b47cee65, public repository `main` contains the exact problem predicate, finite exclusion through 4×10^9 , the window-pair closure compiler, and the V14 Vaughan structural spine. The public V14 status explicitly leaves `AFFINE287-PRIME-MODULUS-MU-TWOOUTER45`, the $k = 0$ smooth packet, fixed-certificate leakage, Gate closure, and the large- M supply open. The accompanying V14 report records a successful `lake build` of 8111 jobs and no Lean-code use of `sorry`, `admit`, a user axiom, `opaque`, `unsafe`, `native_decide`, or `@[implemented_by]`.

8.2 Midpoint V15 archive

The supplied `midway.gz` archive contains 92 Lean files, including the $\Lambda = \mu * \log$ and balanced-seven modules and a V15 status file. A static audit found no Lean-code occurrences of the prohibited declarations above, and the two analytic interfaces are deliberately uninhabited. However:

- those V15 files were not yet present on public `main` at the audit cutoff;
- the archive's top-level build summary records the completed V14 replay, while the user reported that the V15 Lean run was still in progress;
- the present execution environment did not contain `lake`, so no independent local V15 kernel replay was performed.

V16 therefore does not promote the V15 formalization to `PUBLIC LEAN-CHECKED`. Once the ongoing replay completes and is merged, the formal-status table can be promoted without changing the mathematical exposition.

8.3 Document organization

The previous V15 PDF is preserved as an immutable technical patch-stack archive. This V16 document is the coherent public paper; hostile audits, failed provider dictionaries, historical residuals, Gate-specific archaeology, and full chronological provenance live in the separate technical verification ledger. The clean source is compiled reproducibly in the public repository, and the legacy forum PDF path is retained as an alias to the current clean paper.

9 Conclusion

The strongest unconditional result presently available in this project is finite: Erdős Problem #287 has no counterexample with maximum denominator at most 4×10^9 . The exact remaining global input is also clean: prove $\text{WindowPairSupply}(M)$ for every sufficiently large M with an effective threshold inside the verified finite range.

The recent analytic work has reduced parts of the source without claiming cancellation. The identity $\Lambda = \mu * \log$, the hard affine determinant-one line, and the balanced-seven polarization are now exact algebraic infrastructure with V15 formalization source present and final public replay pending. The next mathematics is not another document-level reorganization: it is a genuine signed prime-distribution estimate and a source-matched comparison theorem. Until those are proved and compiled into a large- M supply, Erdős Problem #287 remains open.

A Formal theorem and source map

Public Lean bank at the V16 cutoff

- **Exact problem predicate and public ordered form.** `RequestProject/Erdos287/ProblemStatement.lean`; declarations `Erdos287Counterexample` and `erdos287_seq_of_no_counterexample`.
- **Finite blocker and verified range.** `Gap2CE.holes_isolated`, `Gap2CE.blocker_window`, and `no_Erdos287Counterexample_of_max_le_4e9` in the finite-range files.
- **Large- M conditional compiler.** `WindowPairSupply` and `no_Erdos287Counterexample_of_closure` in `RequestProject/Erdos287/ClosureInputs.lean`.
- **Public analytic-source status through V14.** `Erdos287VaughanV14Status.lean` in the public status directory; exact structural spine banked, analytic child explicitly open.

Midpoint V15 formalization source – public replay pending

- `AffineMuLogIdentity.lean`: exact $\Lambda = \mu * \log$ identity and affine specialization.
- `AffineMuLogHardSource.lean` and `AffineMuLogLine.lean`: exact small- q /small- r /hard partition and determinant-one line.
- `BalancedSevenFinite.lean` and `BalancedSevenPolarization.lean`: coefficient -20 , finite collision/exponent routers, and labelled polarization.
- `PolarizedOmega7SignedEoD` and `MuLogComparisonLowCondMatch`: deliberately uninhabited open specifications, not analytic proofs.

References

- [1] P. Erdős and R. L. Graham, *Old and New Problems and Results in Combinatorial Number Theory*, Monographies de L'Enseignement Mathématique, 1980, p. 33.

- [2] T. F. Bloom, *Erdős Problem #287*, Erdős Problems archive, problem provenance and status page, accessed August 2026.
- [3] K. Ford and J. Maynard, *On the theory of prime producing sieves*, arXiv:2407.14368, 2024.
- [4] S. Drappeau, A. Granville and X. Shao, *Smooth-supported multiplicative functions in arithmetic progressions beyond the $x^{1/2}$ -barrier*, *Mathematika* 63 (2017), 895–918.
- [5] A. Pascadi, *On the exponents of distribution of primes and smooth numbers*, arXiv:2505.00653, 2025.
- [6] The Lean and Mathlib communities, *Lean 4 and Mathlib*, formal proof assistant and mathematical library.